

Nan Jiang

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Education

Data Scientist Certification - Fellowship Program, The Data Incubator

Fordham University, Bronx, NY

Ph.D. in Financial Economics

G.P.A: 3.97/4.0

Valparaiso University, Valparaiso, IN

M.S. in International Economics and Finance

G.P.A: 3.68/4.0

Shanghai University, Shanghai, China

B.S. in Finance

Research Interests

Financial Economics, Applied Econometrics, Machine Learning, Information Fusion, Ensemble Learning, Contract Theory

Working Papers and Projects

Enhancing ADMET Property Models Performance through Combinatorial Fusion Analysis ([Github](#), [Paper](#), [TDC Ranking](#))

Research Methods: Use information fusion of machine learning models and three drug encoding schemes (Morgan FP, RDKit 2D descriptors, and MCFP, a proprietary encoding scheme) to predict properties of ADMET benchmark group data sets.

Result: #1 in the leaderboards in 4 out of 22 datasets, #2 – 6 in 12 out of 22 datasets, and #7 and over in 6 datasets. **Submitted to JCIM in Dec 2023.**

Equity Ranking with Stochastic Dominance and Combinatorial Fusion ([Github](#), [Web app](#))

Research Methods: Use information fusion of machine learning models and nonparametric stochastic dominance of four orders to make ranking predictions of U.S. stocks.

Result: The constructed long-only portfolio outperforms the S&P 500 in both cumulative returns (89.5% vs 26.5%) and Sharpe (1.4 vs 0.7) significantly on July 1, 2020 – July 1, 2022. The web app of stock ranking has been updated actively.

Multitasking Model with Career Concerns

Research Methods: Extension of the multitasking model in Benabou and Tirole (2016) with career concerns.

Result: The introduction of career concerns into the multitasking model motivates the monopsony employer to raise bonus rates for both high type and low type workers, while it has no impact on the employers in the perfect competition labor market, the low type is properly incentivized, and the high type is over-incentivized.

Model of Strategic Voting: An Extension of Voting Behavior

Research Methods: A simple two-candidate election model of voting under plurality rule.

Result: To maximize a voter's expected utility gain, an agent would misrepresent herself in some certain political issues instead revealing her true positions.

Industry Work Experience

OBEX Securities, New Rochelle, NY

Quantitative Researcher

Jun 2023 - present

- Designed, developed, and maintained quantitative stock ranking models and trading strategies using machine learning and information fusion techniques in Python and R.
- Built a unified Python-based risk management system for algorithmic trading with real-time monitoring, VIX-adjusted exposure limits, ATR-based stop losses, HHI-based concentration analysis, and urgency-driven trade execution. Integrated dry-run mode, market regime detection, and price validation to ensure robust, fail-safe operations.
- Programmed custom Python functions for data preprocessing and calculating

model diversity and performance strength, which served as feature weights for ensemble modeling. Employed machine learning libraries such as Scikit-learn and XGBoost, using randomized train-test splits; wrote hundreds of lines of clean, modular code.

- Retrieved and cleaned financial data using Refinitiv Workspace's CodeBook API; conducted local model training and prediction to generate ranking outputs and prepare order import files for VWAP execution across various trading platforms.
- Implemented and executed basket algorithmic trades for both long-only and long-short strategies in separately managed accounts (SMAs) at the start of each month. Achieved a 30% return (Nov 1, 2023 – Sep 30, 2024) for the long-only strategy and a 17% return with a beta of 0.04 (Jan 2 – Sep 30, 2024) for the long-short strategy.
- Led weekly strategy meetings focused on technical improvements and performance analysis.
- Maintained and updated the strategy brochure, and adjusted portfolio allocations based on the latest model predictions and training results.

Financial Analyst

Oct 2017 – Aug 2019

- Monitored daily balances of accrued interest from equity and fixed income financing, hard-to-borrow lending, execution commissions, and fixed income custody markups and fees; responsible for monthly reconciliations and settlements.
- Tracked and updated proprietary trading positions and profit & loss (P&L) using mark-to-market valuation methodology.
- Evaluated the performance of the firm's quantitative options trading strategies by analyzing market trends and indicator movements; supported option and equity traders in daily trade execution, portfolio exposure management, and delta risk hedging.

Haitong Securities, Headquarter, Shanghai, China

Apr 2012 – Jul 2015

Senior Associate – Fixed Income Underwriting

- Secured a total of \$276 million in funding for clients through various fixed income products.
- Developed bond financing proposals and crafted customized financing solutions to meet clients' specific needs, focusing on debt structure optimization and business expansion.
- Conducted comprehensive due diligence, including analysis of business models, financial health, internal management structures, repayment plans, and operational compliance. Collected and organized supporting documents, culminating in detailed due diligence reports with risk assessments.
- Managed the underwriting and issuance of a wide range of fixed income instruments—including enterprise bonds, publicly and privately placed corporate bonds, financial bonds, medium-term notes, commercial paper, and asset-backed securities (ABS). Coordinated with investors and submitted issuance applications to regulatory authorities such as NDRC, NAFMII, and CSRC.
- Led distribution efforts for fixed income products and organized investor roadshows to support successful placement.

Da'gong Global Credit Rating Co., Ltd. Shanghai, China

Jul 2011 – Jan 2012

Credit Analyst

- Conducted due diligence and collected financial documents for credit analysis.
- Interviewed clients' financial officers to gather necessary information.
- Wrote comprehensive credit rating reports on clients' financial status and future potential.
- Developed data mining skills to analyze specific industries for financial insights.

Teaching Experience

Instructor, Department of Economics, Fordham University

Sep 2021 – May 2023

- Basic Microeconomics (Undergraduate level)

Teaching Assistant, Department of Economics, Fordham University

Sep 2020 – May 2021

- Mathematical Methods in Economics II (Ph.D. core)
- Macroeconomic Theory II (Ph.D. core)

Economics Tutor, Department of Economics, Fordham University

Sep 2019 – May 2020

- Statistics for Economics (Undergraduate level)
- Principles of Microeconomics (Undergraduate level)
- Introduction to Macroeconomics (Undergraduate level)

Journal Refereeing

Journal of Quantitative Economics

Presentations

Fordham-IBM Workshop on Generative AI

Topic: Generative AI and Multi-Property Drug Discovery Prediction

Fordham University
Oct 27, 2023

CISC 6910 – Data and Information Fusion

Topic: Equity Ranking with Stochastic Dominance and Combinatorial Fusion

Fordham University
Apr 17, 2023

Finance Workshop, Department of Economics

Topic: Equity Ranking with Stochastic Dominance and Combinatorial Fusion

Fordham University
Nov 30, 2022

CISC 5900 – Information Fusion

Topic: Equity Ranking with Stochastic Dominance and Combinatorial Fusion

Fordham University
Jun 30, 2022

CISC 5900 – Information Fusion

Topic: Equity Ranking with Stochastic Dominance and Combinatorial Fusion

Fordham University
May 16, 2022

Awards and Honors

2022 - 2023: GSAS Teaching Fellowship, Fordham University

2022: GSAS Summer Research Assistantship, Fordham University

2022: Summer Research Fellowship, Fordham University

2021 - 2022: GSAS Teaching Fellowship, Fordham University

2020 - 2021: GSAS Graduate Teaching Assistantship, Fordham University

2019 - 2020: GSAS Graduate Assistantship, Fordham University

Skills

Software: Python | SQL | R | MATLAB | Stata | LaTeX | Refinitiv Workspace API + CodeBook | Interactive Brokers TWS | JPMorgan Neovest

Date science & machine learning: Information Fusion | Ensemble Learning | Pandas | NumPy | Scikit-Learn | TDC & RDKit | TensorFlow | Matplotlib | Streamlit | Time-Series Analysis | Data Visualization | Data Scrubbing | Spark | NLP | Supervised & Unsupervised Learning | Generative AI

Language: English (Fluent) | Chinese (Native)

References

Prishkesh D. Vinod (Advisor)

Professor

Department of Economics

Fordham University

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✉: vinod@fordham.edu

Erick W. Rengifo

Professor

Department of Economics

Fordham University

☎: 718-817-4061

✉: rengifomina@fordham.edu

Frank Hsu

Clavius Distinguished Professor

Department of Computer and Information Sciences

Fordham University

☎: 718-817-4483

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